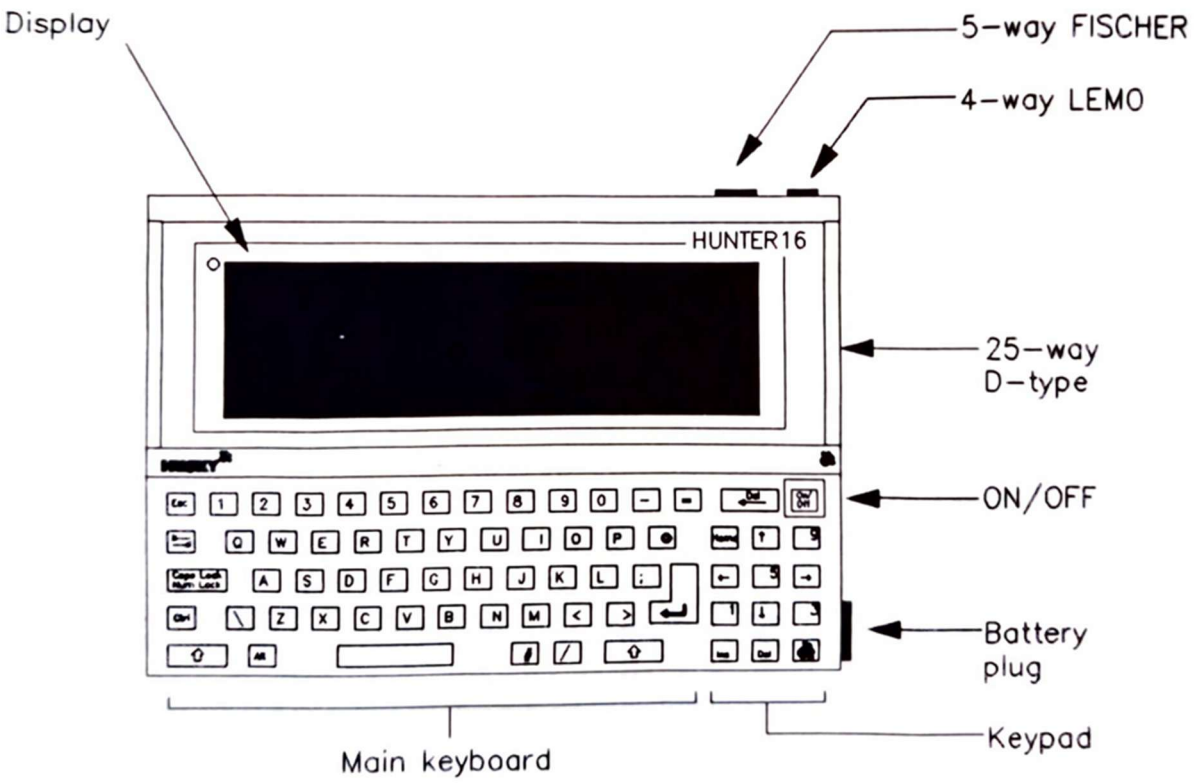
Chapter 5 Using Hunter 16

This chapter briefly describes how to operate Hunter 16: how to turn on, turn off, and reboot; how to use its main controls; and how to use MS-DOS and applications.

5.1 The Hunter 16

The main parts of **Hunter 16** look like this:



[PWR]	used to turn on or off
[PAW]	used to control the screen and other special facilities, and to provide certain IBM-compatible keys
Keyboard	used to type commands and text, and to control **Hunter 16**
LCD screen	displays text and graphics output

D-type connector	RS232 interface for connecting to another computer, modem, serial printer, AC adaptor, etc
S-way FISCHER connector	for connecting the **Oracle GT** disk drive. AC adaptor, a barcode
4-way LEMO connector	for connecting the mains adaptor, barcode wand, or simple RS232 input
battery plug	for inserting and removing batteries

5.2 Turning on

To turn **Hunter 16** on (or off), press the [PWR]) key firmly.

Normally, when you turn on, **Hunter 16** will continue with what it was doing when turned off, almost as if you had not turned it off! For example, if you were running an MS-DOS application, the screen will be restored and you can carry on using it.

This will save you a lot of time and trouble, as you don't need to save the data files your application is sing before turning off, or to load the application and data files again when you turn on.

Note that this is quite unlike a conventional desktop computer, which reboots each time it is turned on. (See section 5.4 for instructions on rebooting.)

If you see a screen like this when you turn on:

This is the "Welcome' screen, which shows that **Hunter 16** is rebooting (for example, because this is the first time it has been turned on). Press any of the 'normal keys (such as the spacebar or [RET]) and **Hunter 16** will perform a normal MS-DOS boot. This is also known as a 'cold start'.

Depending on how your **Hunter 16** is set up, this may start your application automatically, or take you to MS-DOS command mode, for you to use MS-DOS commands or run your applications by typing the appropriate commands.

If you cannot see anything on the screen, you may need to adjust the screen contrast (viewing angle): hold down the [PAW] key and press [X] or [E] until the screen is clearly visible.

If this does not help, the batteries may be run down, or there may be something seriously wrong with **Hunter 16**: see Chapter 7.

NOTE: If **Hunter 16** is operating from batteries, always turn off if you are not going to use it for a while, even for a few minutes. This will help ensure the maximum battery life.

5.3 Turning off

To turn **Hunter 16** off, press the PWR) key firmly. You should turn off whenever you are not going to be using **Hunter 16**, even if it's just for a few minutes, as this will extend battery life considerably.

Turning **Hunter 16** off is not like turning off a desktop computer, because when you turn on again, it will normally continue operating exactly as if you hadn't turned it off! For example, if you turn off in the middle of editing a document, when you turn on again, the editor will be loaded with the document, with the cursor in the right place, etc.

Hunter 16 will also turn itself off, if:

- the battery power is low

or you haven't touched any of the keys for a while (normally five minutes) and **Hunter 16** is not using its serial ports

or an application is running which tells it to turn off

Again, you can just turn on again and carry on working as if nothing had happened, when you want.

An application may stop you turning off by disabling the [PWR] key. For example, it might do this to make sure that it has finished transferring a file via the RS232 port. A message should be displayed to tell you what is happening. If you MUST turn off, see Chapter 7 for advice.

5.4 Rebooting

As already mentioned, **Hunter 16** does not reboot when it is turned on, but continues execution from the point where it was turned off. This makes operation very convenient, but there will be times when you want to reboot. For example, if you change C:\CONFIG.SYS (eg add a device driver), you must reboot to bring the changes into effect.

To reboot, simply use the usual MS-DOS 3-key reset ie press [Ctrl][Alt][DEL]. This displays the 'Welcome' screen:

Press any of the 'normal' keys (such as the spacebar or [RET]) and **Hunter 16** will perform a normal MS-DOS boot, configuring the system as specified by the CONFIG.SYS file and executing the commands in the AUTOEXEC.BAT file. (These files are held in the root directory of the disk drive used to boot; usually drive C (the RAM disk), but if an **Oracle GT**

disk drive is connected and contains a diskette, it will be used instead. To boot from drive C, simply remove the diskette before rebooting.)

Depending on how your **Hunter 16** is set up, this may start your application automatically, or take you to MS-DOS command mode, for you to use MS-DOS commands or run your applications by typing the appropriate commands. (If you want to leave the application, you should have been provided with instructions on how to do this. If not, see Chapter 7.)

5.5 The screen

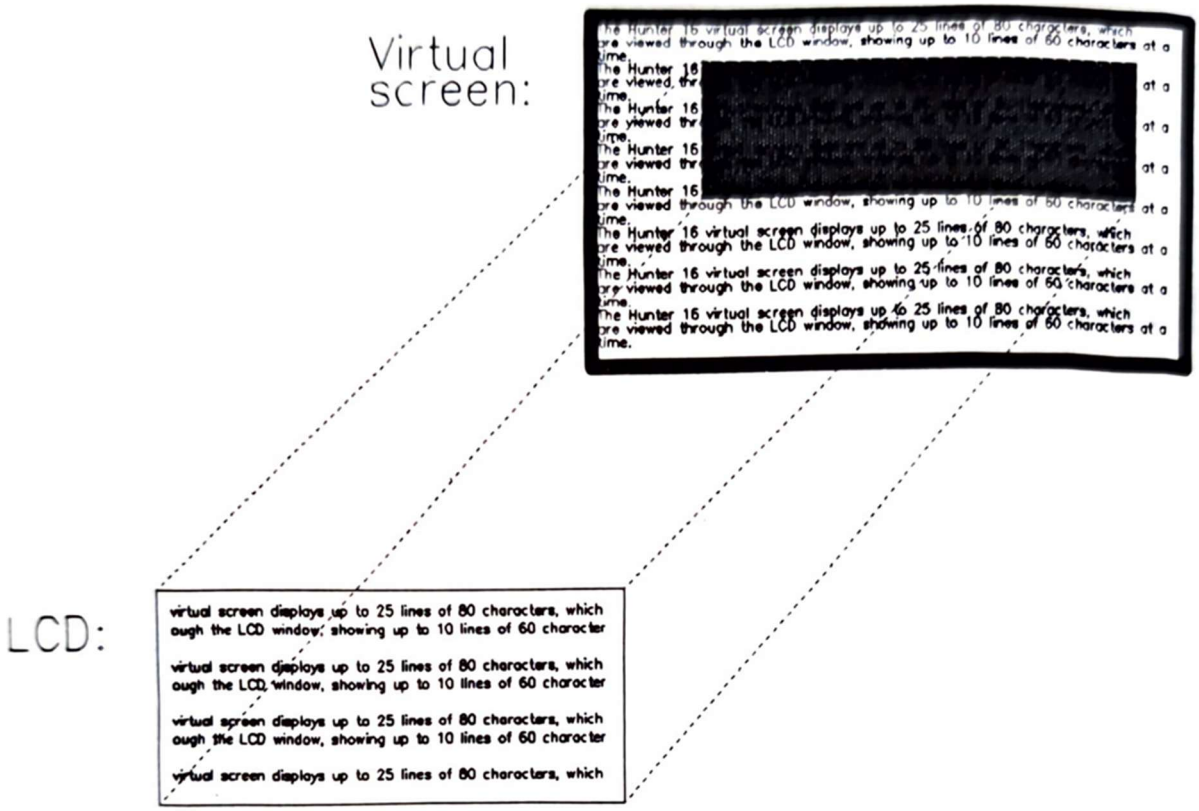
5.5.1 Introduction

Hunter 16 has a large LCD (liquid crystal display) screen, which can show up to 10 lines of 60 characters and pictures using a grid of 240 by 64 dots.

The screen is protected by a polycarbonate window, as strong as glass. Like glass, it can be scratched, so please take care. The window should only be cleaned with a chamois or lens cleaning cloth.

Because it works by contrast, rather than emitting its own light, it is perfectly visible no matter how bright the light is. But a word of caution: try to avoid leaving **Hunter 16** exposed to direct sunlight for prolonged periods-this can damage the delicate chemicals in the LCD.

The screen behaves as a window' onto a larger virtual' screen, which emulates the IBM CGA (colour graphics display adaptor) ie provides 25 lines of 40 or 80 characters (text modes) or 320x200 or 640x200 pixels (graphics modes). (For more detailed information, see Part 3.)



Applications can use the display exactly as if it were a full size IBM CGA screen. The LCD screen 'window' moves over the virtual screen automatically, to show the most recent output (eg a prompt waiting for input, or your typing). You may also control it by using special keys, as follows.

Hunter 16's LCD screen gives good results in most lighting conditions. For use in poor light, there is a backlight.

To adjust the LCD screen for maximum clarity, place it at the angle you will be working, hold down [PAW] and press the [X] or [E] key until it appears clearest. (These are the decrease and increase contrast keys.)

To improve visibility in poor lighting conditions, turn on the backlight: hold down the [PAW] key and press [L]. To turn it off, press the same keys again. The backlight places an additional drain on the batteries, so only use it for short periods.

To maintain legibility, wipe the screen with a lens cleaning cloth or damp chamois leather: do not use solvents as they might damage the screen. To ensure the long and trouble- free service life of which it is capable, do not leave the screen in bright sunlight for long periods.

5.5.3 Moving around the virtual screen

The virtual screen provides a CGA-compatible screen, allowing up to 25 lines of 80 characters and graphics with resolutions up to 640x200 pixels. The LCD provides a window onto this virtual screen, which can display up to 10 lines of 60 characters, or 240x64 pixels.

The window moves over the virtual screen automatically to show the most recent keyboard input. You may also move the window by pressing the 'viewing keys':

To flip' the window to show the other side of the virtual screen, hold down the [PAW] key and press [F]. (This is the LCD flip key.)

To move the window a line or column at a time, hold down [PAW] and press one of the cursor keys. (These are the LCD looking keys.)

The window will return to its original position when you type text or hold down [PAW] and press the [5] key on the numeric keypad. (This is the LCD home key.)

5.5.4 Screen lookup (zoom) key

To display the contents of the full screen, hold down [PAW] and press [M]. This displays a pixel on the LCD screen for each nonspace character (if in text mode) or group of 4x4 pixels which are set to black (if in graphics mode).

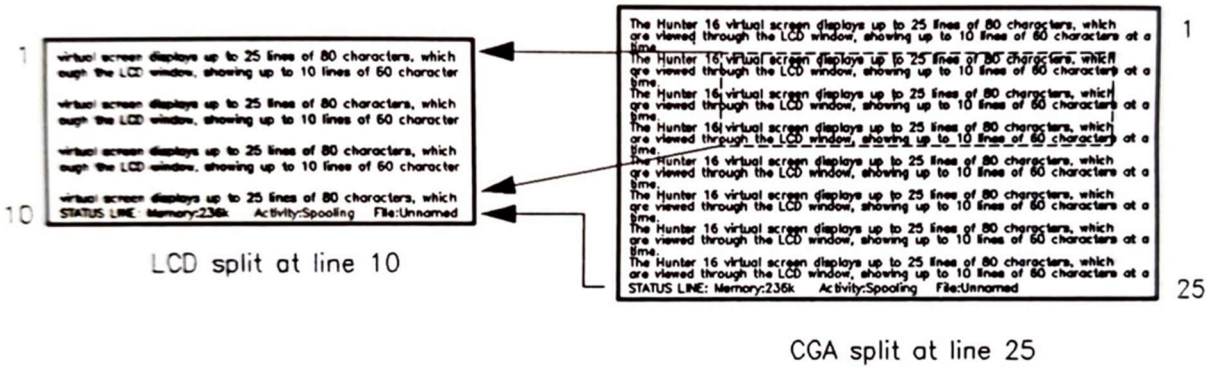
To return to the normal LCD contents, press the same keys again.

5.5.5 Split screen

Many applications use a particular screen line (usually the top or bottom line) for status information and use the rest for the information which is being entered or displayed. As the LCD screen shows whichever part of the full screen is being updated, this could lead to a lot of distracting screen movement.

To help prevent this, you can split the virtual screen into two parts. and display each in a separate part of the LCD screen. Each part of the LCD can either be fixed in position or scrolled to show output in the corresponding part of the virtual screen.

For example, if your application uses line 25 as a status line, you could split the virtual screen at line 25, and split the LCD screen at line 10:

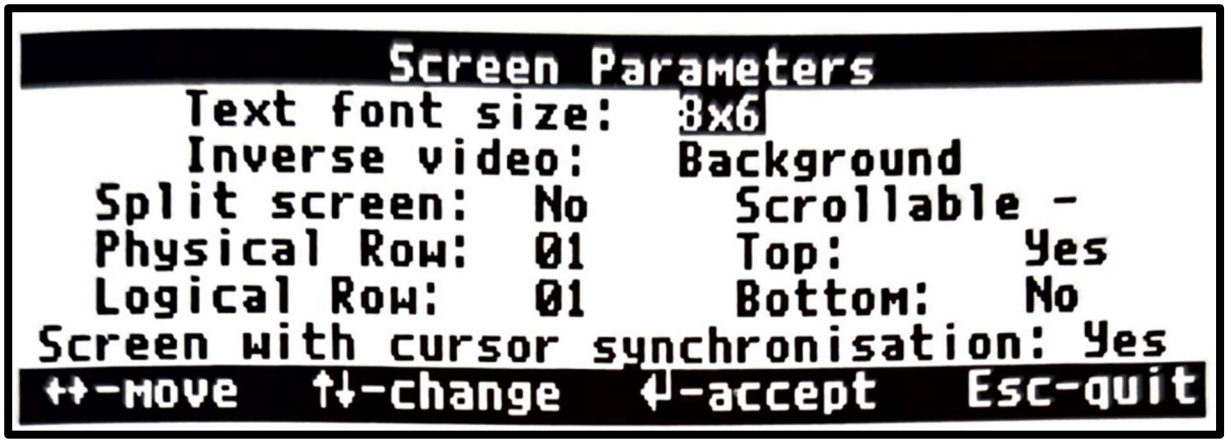


The top nine lines of the LCD could be set to scroll (to show data input/output), while the bottom line could be set to fixed (to show the status line).

To do this, press the HOT key ([PAW][H]), and you'll see a screen like this:

The last line of the screen indicates the keys you can use. The current selection (`Screen`) is highlighted by reversing it.

Use the [Left] or [Right] keys to move the highlight to the `Screen` option, then press [Return] to select it. You'll see a screen like this:



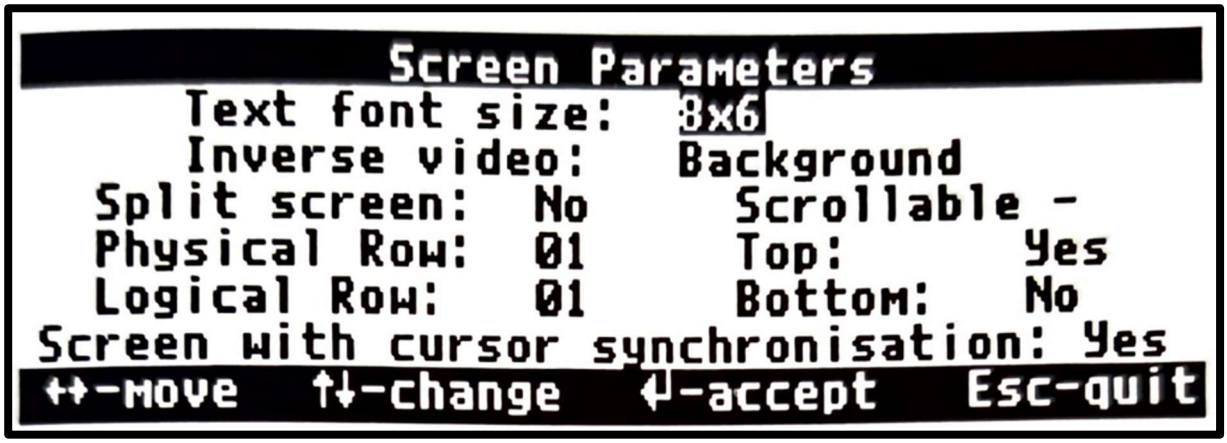
Move the highlight to the options you want to change by using, the [Left] and [Right] keys, then adjust the display value by pressing [Up] or [Down] key. The 'physical row' is the row at which to split the LCD. The ‘logical row’ is the row at which to split the CGA. When the screen is split as you want it, press [RETURN], then press [Esc]) to leave the HOT key.

5.5.6 Inverse video

An IBM CGA Screen (which **Hunter 16** emulates) can display text with a wide range of attributes, including foreground (text) colour (red/green/blue) and intensity, background colour (red/green/blue) and blinking. These attributes are typically used to highlight options or important messages, or for purely decorative purposes.

The **Hunter 16** screen is monochrome, so it obviously can't support colour! It can however display text as reverse video (white on a black background) and you can choose to use this to emulate some of the other text attributes.

To change the way reverse video is used, press the HOT key ([PAW][H]), make sure the highlight is on the `Screen` option (if not, use [Left] or [Right] to put it there), then press [RETURN]. You'll see a screen like this:



Move the highlight to the **`Inverse`** `video` option with the [Left] and [Right] keys, then adjust the displayed value by pressing [Up] or [Down] key. The options are as follows:

`Background`	Use inverse video for text with a background colour other than black.
`Intensity`	Use inverse video for high intensity (bright) text.
`High Back`	Use inverse video for text with a high intensity (bright) background.
`Int/Backgd`	Use inverse video for text with a high intensity (bright) text or background.
`None`	Do not use inverse video.
`Monochrome`	Use inverse video for text with a black foreground and a white background (as on an IBM monochrome display adaptor).

When you've set inverse video as you want it, press [RETURN], then press [ESC] to leave the HOT key.

If you have problems reading the screen, experiment with changing the inverse video method used to get the best effect.

5.6 The keyboard

5.6.1 Introduction

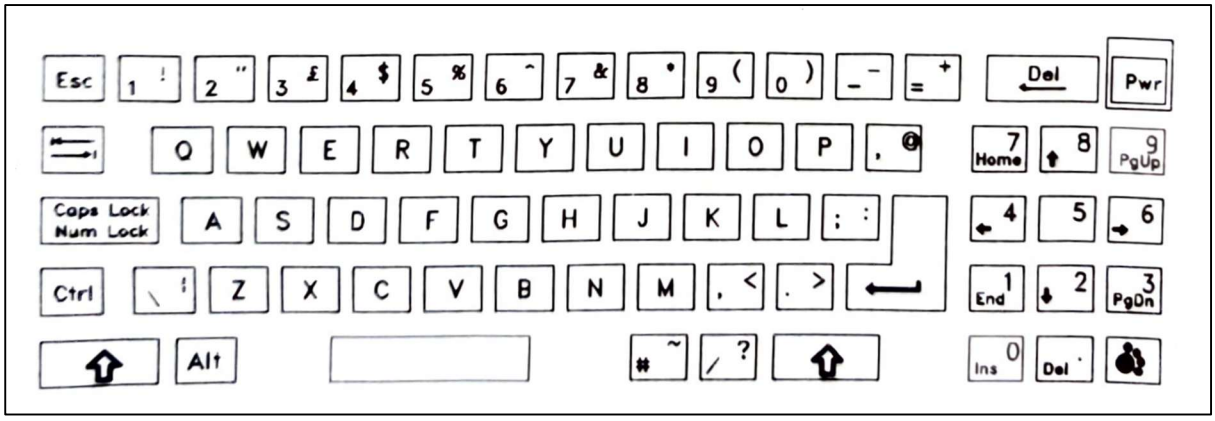
The **Hunter 16** keyboard is used for two functions: for typing and to control **Hunter 16**.

Typing is provided by keys organised into a standard QWERTY layout, with a separate numeric keypad/cursor key cluster. The keyboard provides all the functions that you are likely to need in an MS-DOS environment, but some of the keys are in different positions to an IBM keyboard, and some keystrokes are implemented as key combinations rather than separate keys.

The control functions are implemented through an extra 'shift key, the [PAW] key. These functions are used to control the display, backlight, sound and other facilities.

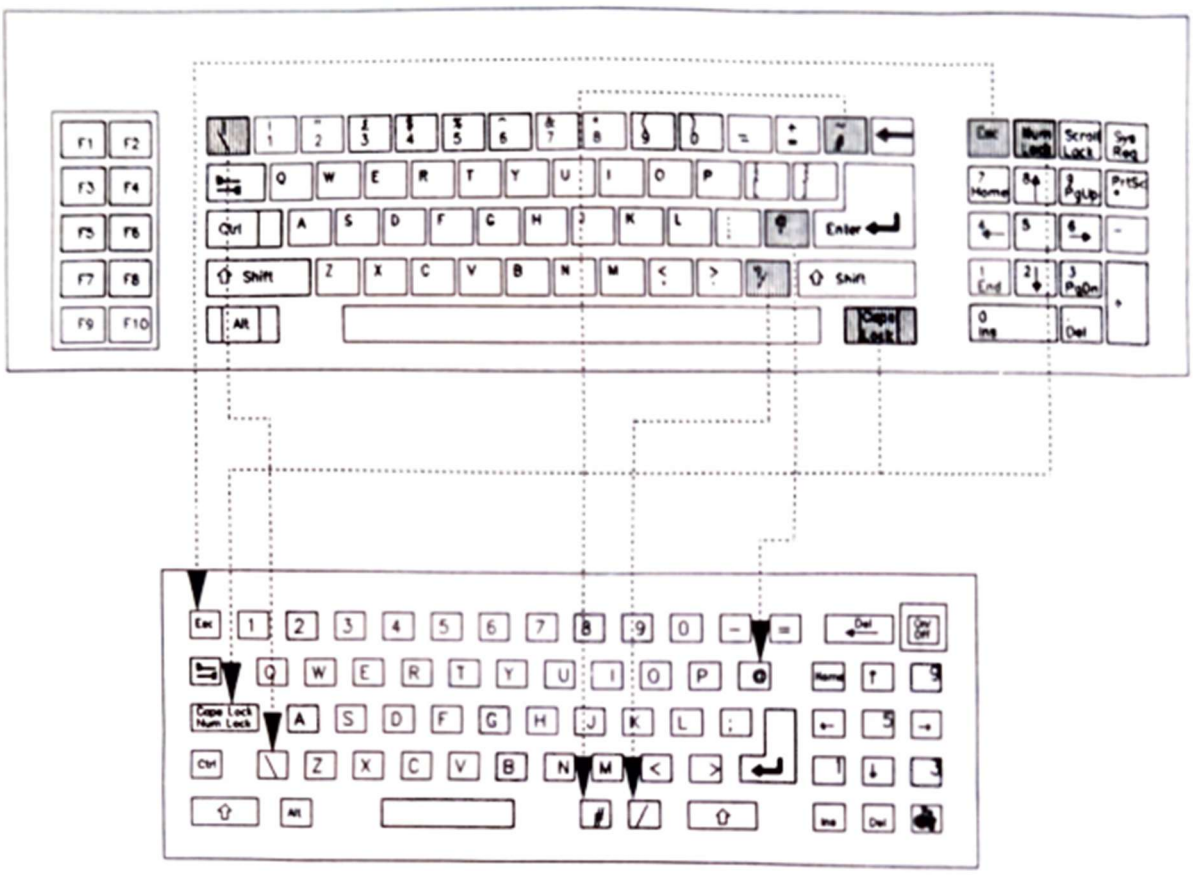
5.6.2 Typing keys

The standard keyboard looks like this:

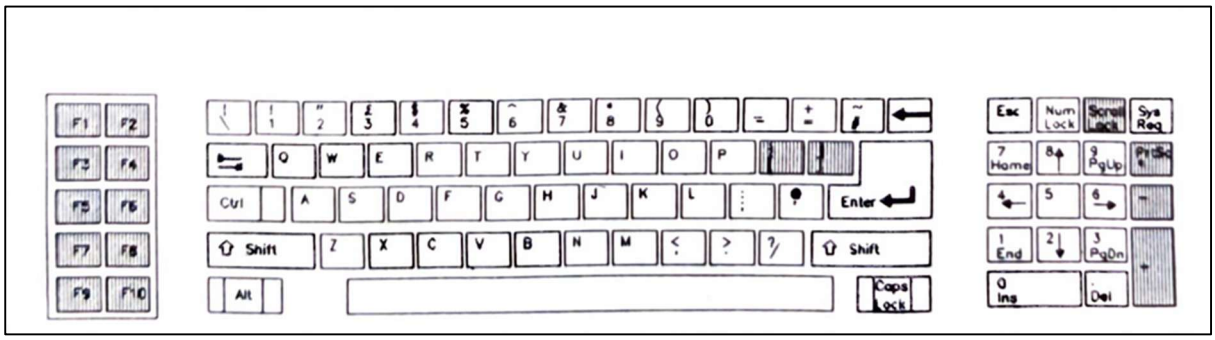


Other keyboard variants are also available to cater for national differences.

The keyboard provides all the keys you'll need in an MS-DOS environment, but to provide portability, the layout is a little different to a desktop IBM-compatible. Some keys have simply been moved:



While certain other keys must be typed as key combinations, using the special [PAW] key:



For:	Type:
F1 F2 ...F10	plus the function key number, from the top row of the keyboard (not the keypad). Use [0] for E10.

Scroll Lock	[PAW] plus [9] from the keypad
Ctrl Break	[Ctrl][PAW] plus [9] from the keypad
Prtsc	[Shift][PAW] plus [7] from the keypad
[	[PAW] plus [1] from the keypad
]	[PAW] plus [3] from the keypad
{	[Shift][PAW] plus [1] from the keypad
}	[Shift][PAW] plus [3] from the keypad
Gray -+	[PAW][#]
Gray --	[PAW][/]

The [PAW] key is also used to provide a number of **Hunter 16**-specific control functions, as described in the next section.

5.6.3 Special Keys

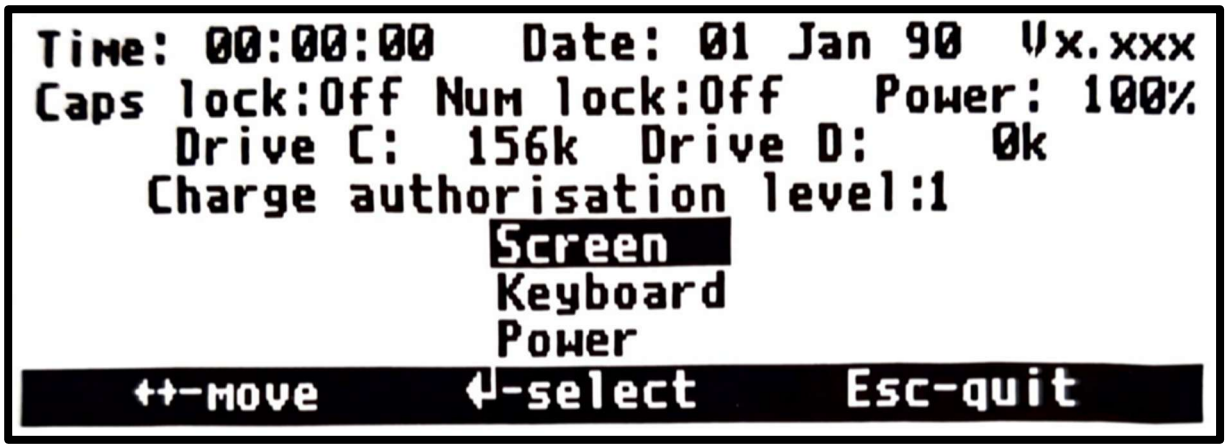
The [PAW] key also provides access to a number of control functions, as follows:

To:	**Type:**
Shift window left	[PAW][Left]
Shift window right	[PAW][Right]
Shift window up	[PAW][Up]
Shift window down	[PAW][Down]
Flip window	[PAW][F]
Restore window position	[PAW] and [5] on the keypad
Increase contrast	[PAW][E]
Decrease contrast	[PAW][X]
Display LCD map	[PAW][M]
Turn backlight on/off	[PAW][L]
Increase backlight level	[PAW][R]
Decrease backlight level	[PAW][C

Turn sound on/off	[PAW][S]
HOT key	[PAW][H]

The screen control keys have already been described in section 5.5.

The HOT (or status) key is used to display and optionally alter aspects of the **Hunter 16's** set-up. When you press this key combination, the screen displays the following information:



- The time and date
- The state of the [Caps Lock] and [Num Lock] keys
- If Advanced Power Mode (APM) is enabled, the percentage power remaining and charge authorisation level. (This dictates whether you can recharge the internal batteries by using the **Husky** AC adaptor. For details, see section 6.5.5.)
- The free space remaining on drive C (RAM disk) and drive D.

To return to the normal screen display, press [Esc] or press [PAW][H] again.

This key also allows you to adjust the screen, keyboard and power settings. For full details, see Chapter 9.

5.6.4 International keyboards

While the standard keyboard layout is designed for the United Kingdom, other keyboards may also be ordered to cater for other languages. (For details, see Appendix E.)

To inform the **Hunter 16** that you are using a different keyboard layout you must use the INTKEYB utility, which is stored in the root directory of drive C. Note that you should NOT just use the standard DOS KEYB utility, as this will not modify the extended BIOS information.

To switch to another keyboard layout, type INTKEYB followed by the appropriate switch: /F (for French), /G (for German), /S (for Swedish), or /U (for US American). For example, to select a German keyboard, type:

```
INTKEYB /G [RETURN]
```

This loads the relevant keyboard driver using the KEYB command, then modifies the extended BIOS information for the new keyboard. To make sure that the correct keyboard layout is always installed, add the INTKEYB command to AUTOEXEC.BAT.

To return to the UK layout, type INTKEYB /R.

5.7 Using MS-DOS

Hunter 16's operating system is the industry-standard for desktop computers, Microsoft's MS-DOS. The version used is 3.3.

If you are not already familiar with using MS-DOS (or the closely related PC-DOS), you should consult the *MS-DOS User's Reference* for an introduction to details of the standard facilities and commands.

The following standard MS-DOS commands will always be available, whatever directory you are in (as they are built into C:\COMMAND.COM):

BREAK	Turn detection of [Ctrl][Break] on or off
CD (CHDIR)	Change directory
CLS	Clear screen
COPY	Copy file(s)
DATE	Set/display date
DEL (ERASE)	Delete files
DIR	Display directory/file details
MD (MKDIR)	Create directory
PATH	Define search path for commands
PROMPT	Redefine system prompt
REN (RENAME)	Rename files
RD (RMDIR)	Delete directory
SET	Set environment variables
TIME	Set/display time
TYPE	Display contents of text file
VER	Display MS-DOS version number
VERIFY	Turn verification of disk writes on or off
VOL	Display disk label

The GWBASIC BASIC interpreter is also supplied, together with the following MS-DOS utilities, which are in the C:\DOS directory:

ANSILSYS	FIND.EXE	PRINT.COM
APPEND.EXE	FORMAT.COM	REPLACE.EXE
ASSIGN.COM	GRAPHICS.COM	RESTORE.COM
ATTRIB.EXE	GWBASIC.EXE	SORT.EXE
BACKUP.COM	JOIN.EXE	SUBST.EXE
CHKDSK.COM	KEYB.COM	SYS.COM
COUNTRY.SYS	KEYBOARD.SYS	TREE.COM
DEBUG.COM	LABEL.COM	XCOPY.EXE
FASTOPEN.EXE	MODE.COM	
FC.EXE	MORE.COM	

For details of GWBASIC, see the *GWBASIC User's Guide*. For details of the MS-DOS utilities, see the *MS-DOS User's Reference*.

These are supplied as (read-only) files in the C:\DOS directory. If you find you can't use these utilities, add this directory to your search path. (For details of search paths. see PATH in the *MS-DOS User's Reference*.)

The following Hunter 16-specific utilities are also provided as (read-only) files in the C:\ directory:

COM	Simple file transfer utility
EDIT	General purpose text editor
INTKEYB	International keyboard support
UTIL	Format internal or external disks
	Set time/date
	Configure communications
	Configure memory

Again, if you want to use these utilities, you may need to add their directory (C:\) to your DOS search path.

(You can also use the standard MS-DOS FORMAT command to format diskettes in the **Oracle GT** drive or a desktop computer's drive, but if you want to boot from a diskette, you must prepare it on **Hunter 16**.)

There is also a status/set up function, accessed via a HOT key.

COM, EDIT, INTKEYB, UTIL and the HOT key are all described in detail later in this guide. Other commands, utilities and applications may also have been installed for you to use, by your technical support team. If you are meant to use them, instructions should also be supplied.

5.8 Using applications

If **Hunter 16** has already been set up to run the application(s) that you want automatically, you probably won't need to know anything about MS-DOS or the extra utilities provided.

Refer to the instructions supplied with the application for general guidance on using it. If the application has not been produced or adapted for use on **Hunter 16**, you will need to modify the instructions slightly to take account of the different keyboard and screen, as described earlier in this chapter.